Health-a-thon 2026

FINAL SUBMISSION

[Short subtitle explaining what your solution does]

BEAVERS

TEAM CODE [Insert team code]

Team Overview & Background

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Problem Statement and Supporting Evidence

01 Excessive Documentation & Manual Record Review

Clinicians spend significant time navigating records with different standard unit of measurement, reconstructing patient histories, and preparing medical documentation.
According to a U.S. observational study, physician spend 49.2% [1] of their time in the EHR and desk work

[1] Sinsky et al. — Allocation of Physician Time in Ambulatory Practice

02 Clinician Burnout & Cognitive Overload

Clinicians face increasing mental and physical strain from repetitive documentation, prolonged EHR interactions, and constant switching between fragmented medical records. The need to manually reconstruct patient histories and identify critical information adds to cognitive fatigue,and can divert attention from direct patient care.

03 Consultation Amnesia & Medication Confusion

During consultations, expectant mothers struggle to retain verbal instructions, leading to critical advice going unheeded and confusion over medication dosages. According to a landmark clinical review, 40% to 80% [2] of medical information provided orally by doctors is forgotten immediately, with nearly half recalled incorrectly.

[2] Kessels R.P.C. — Patients' Memory for Medical Information (Journal of the Royal Society of Medicine, 2003

04 The Inter-Visit Surveillance Void & Adherence Collapse

While 96% of mothers receive paper MCP cards and 88% receive IFA tablets, physical records remain unindexed and inter-visit compliance drops drastically: only 44% take IFA for at least 100 days, and 41% fail to receive the WHO-recommended 4 ANC visits. Between monthly clinic visits, clinicians have zero visibility into daily physical activity, vitals, or emerging red flags like creeping blood pressure

[3] NFHS-5 (2019–21), MoHFW / IIPS — Chapter 8: Maternal Health

05 Maternal Isolation & Perinatal Mental Health Gap

Expectant mothers lack safe, structured peer-support community spaces to share lived experiences, voice fears, and validate day-to-day pregnancy challenges. Perinatal depression affects 15% to over 30% of pregnant women in developing regions; without continuous inter-visit mood tracking and peer reassurance, acute distress goes unmonitored and stigmatized until reaching a clinical breaking point.

[4].WHO Maternal Mental Health Framework & The Lancet Psychiatry Perinatal Peer-Support Trials (THPP)

Target Users and Use Case

Primary Users & Intended Healthcare Setting



01. Expectant Mothers (Patients & Families)

Healthcare Demographics: Rural, semi-urban, and urban pregnant women across Trimesters 1, 2, 3 and Postpartum (covering all literacy levels).

- **Core Needs:** Friction-free tracking (<30s), clear medicine instructions (what, when), and a safe peer community.
- **Pain Relieved:** Eliminates fear of forgetting doctor's advice,confusion over medication schedules, and feelings of maternal loneliness.

02. Obstetricians & Antenatal Nurses (Clinicians)

Healthcare Settings: High-volume public PHCs, District Civil Hospitals, and private maternity clinics handling 60-80+ patients/day.

- **Core Needs**: A 30-second synthesized patient summary and zero-typing (ambient) clinical documentation.
- **Pain Relieved:** Reclaims consultation time from repetitive intake (e.g., LMP, vitals) and removes manual EMR typing, restoring direct patient eye contact.

Workflow Integration (How It Fits)

01 Pre-Visit Intake & Trend Synthesis (Home to Waiting Room)

Self-paced intake: Mother logs daily activity and answers routine queries (e.g., LMP).
- **Lab & paper OCR:** Photos of lab slips and MCP cards are parsed and digitized into trends.
- **30-second doctor brief:** A concise summary with key trends and verbatim patient questions.

02 Ambient Dual-Device Consultation (In Examination Room)
- **Frictionless pairing:** Doctor clicks "Start", patient confirms via 6-digit code (bilingual consent).
- **100% eye contact:** Consultation and physical exam without typing or screen distraction.
- **Instant SOAP note:** Auto-generated structured clinical draft with cited evidence for 1-click sign-off.

03 Automated Post-Visit Care & Medication Reminders (Patient Dispatch)
- **Digital take-home prescription:** Approved notes sent instantly with clear, jargon-free medication schedules.
- **Smart reminders:** Automated daily alerts for essential regimens (e.g., Iron-Folic Acid,Calcium, Antihypertensives).

04 Inter-Visit Surveillance & Peer-Mother Community (The 28-Day Bridge)
- **Continuous micro-check-ins:** 30-second surveys to track symptoms, vitals, and activity between 4-week visits to catch red flags early.
- **Peer community:** Moderated space for expectant mothers to share experiences, normalize anxiety, and proactively prevent perinatal depression.

Solution Overview, Key Features & Impact

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01 Continuous Prenatal Check-ins & Health Connect

- **Three-tier check-ins:** Weekly (gestational-age based), quick symptom pulses, and anytime concern logging.
- **Patient-led reporting:** Symptoms, emotional well-being, medication adherence, and questions for the next visit.
- **Red-flag guidance:** Flagged responses prompt immediate guidance to contact the maternity care team.
- **Wearable integration:** Permission-based import via Android Health Connect (steps, sleep, HR, BP, weight).
- **Daily trends:** Aggregate readings into daily records, synced via clinic systems, and compared over time for clinician review.

02 Interval-Scoped AI Pre-Consultation Brief

- **Visit-to-visit context:** Uses the last visit and all subsequent check-ins, concerns, vitals, and wearable trends.
- **Gemini-powered synthesis:** Priority flags, symptom trajectories, emotional well-being, adherence summary, and suggested discussion points.
- **Patient questions retained:** Carries the mother's questions into the brief.
- **Deterministic fallback:** Rule-based summarizer generates a brief if Gemini is unavailable (system remains accessible).
- **On-demand refresh:** Clinicians can regenerate the brief with new data.

03 Dual-Device Ambient Consultation & Evidence-Linked SOAP Drafts

- **Simultaneous capture:** Doctor's workstation and patient's phone record the consultation after consent.
- **Retryable uploads:** 15–25s audio chunks over HTTPS with a unique session ID, device, sequence ID to prevent duplicates.
- **CrossTalk suppression:** Temporal overlap, token & character-bigram Dice similarity, and relative loudness to remove duplicate utterances.
- **Separate drafting engine:** Gemini or a local model generates structured SOAP drafts with transcript references.
- **Clinician control:** Review, edit, approve, or reject; missing audio and uncertain content remain visible.

04 Lab OCR, Longitudinal Records & Prescription Support

- **Document digitization:** OCR (PaddleOCR) extracts report content.
- **Metric standardization:** Alias matching and unit conversion for comparable results over time.
- **Uncertainty review:** Unmatched or uncertain values flagged for verification.
  Longitudinal context: Lab results, vitals, and maternity-card observations show changes across visits.
- **Saved-template retrieval:** Gower similarity × data coverage to rank reusable prescription templates.
- **Inline completion:** Populate editable medicine, dose, frequency, duration via explicit selection or acceptance.

05 Explainable Hybrid Patient Similarity

- **Algorithm:** hybrid-gower-shrinkage-v2.
- **Combines three sources:**
  - **Structured data:** Custom-weighted Gower similarity (numerical, ordered, categorical).
  - **Symptoms:** Severity-weighted, polarity-aware matching.
  - **Narrative:** Dense embedding cosine similarity (fallback: unigram + bigram).
- **Retrieval pipeline:** Gestational-age filtering, normalization,asymmetric missing-data handling, and Bayesian coverage shrinkage.
- **Explainable results:** Shows matched factors, conflicts, coverage, component scores, and unchecked factors.

$$G^* = \frac{(w_G(G_o + 0.5c) + \kappa S_o)}{W + \kappa} \qquad S = \frac{\alpha G^* + \beta P^* + \gamma N}{\text{sum of active weights}}$$

Default weights: 45% structured, 35% symptoms, 20% narrative.

06 Natural-Language Workflow Automation

- **Conversational navigation:** English, Hindi, and Hinglish requests can open supported pages and records.
- **Role & patient validation:** Server checks permissions and resolves patients within accessible records.
- **Grounded field entry:** Proposed values are verified against the user's request.
- **Constrained LLM planning:** Other requests use an Ollama/ OpenAI-compatible provider that returns a structured JSON plan.
- **Sequential execution:** Opens records, fills controls, and runs supported steps in order.
- **Confirmation before changes:** Saving or sending requires explicit confirmation; re-checks if the form changes.

07 Post-Visit Continuity & Screened Peer Support

- **Accessible care plan:** Revisit consultation notes, prescriptions, and follow-up tasks.
- **Interactive checklists:** Mark assigned actions complete and log concerns between visits.
- **Screened peer discussions:** Tagged feed and threaded replies to share lived experiences.
- **Separated identity:** Capability-based authorization separates community access from clinical identity.
- **Layered screening:** Identifier redaction, configured rules, and local-LLM prioritization.
- **Professional outreach:** Concerning submissions can enter a configured human-review and private outreach workflow.

End-to-End Workflow: Patient & Doctor Journeys

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Patient Journey



01 Sign in & Set Up Profile

Accept the clinic invitation or sign in, complete the patient profile and view upcoming appointments.

02 Complete Check-ins

Submit weekly reviews, quick symptom pulses or anytime concerns. Record well-being, medication adherence and questions for the doctor.

03 Share Health Information

Upload laboratory reports
and sync available Health
Connect readings after granting permission.

04 Join Consultation & Consent

Enter the doctor's room code,review recording consent,allow microphone access and participate in the consultation.

05 View the Care Plan

Access the doctor's reviewed notes, prescriptions and follow-up tasks in visit history.

06 Continue Care at Home

Complete assigned tasks, log new concerns and participate in screened peer discussions.

Doctor Journey



01 Open or Schedule the Patient Visit

Select the patient and appointment, or create a new visit and share the clinic invitation.

02 Review the Pre-Consultation AI Brief

Scan updates since the previous
visit, symptom and wearable
trends, priority flags and the
mother's questions.

03 Verify Records & Review Comparable Cases

Check laboratory values, uncertain OCR results and maternity history. Review patient-similarity matches, conflicts and unchecked factors.

04 Record the Consultation

Create the room code, confirm consent and start dual-device recording. Stop audio capture when the conversation ends so transcript and draft processing can finish.

05 Review Notes & Prepare the Prescription

Edit the transcript-linked SOAP draft. Review medicine choices, saved templates, known allergy flags and applicability conflicts.

06 End the Visit & Share the Care Plan

Confirm the notes, prescriptions and follow-up tasks, then make the visit summary available in the patient's history.

Technology, Tools & AI Approach

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Health-On connects patient and clinician applications through shared processing services, validated APIs and a common care record. AI prepares information for review; clinicians control the final care plan.

System Architecture & Key Technologies



Patient App
Flutter / Dart

- Check-ins & concerns
- Report uploads
- Consultation & summaries
- Follow-up & community

Backend API
Node.js + Express

Processing Services
Python + FastAPI
(OCR, Transcription, AI Tasks)

Shared Platform
Supabase
(Auth, PostgreSQL, RLS, Storage)



Clinical Dashboard
React + Vite + Tailwind

- Appointments & records
- AI pre-visit brief
- Consultation drafts (SOAP)
- Prescriptions & templates

AI Models, Algorithms & What They Enable

Lab report extraction	PaddleOCR, pdfplumber, PyMuPDF	Structured, standardized lab results
Consultation transcription	Sarvam Saaras v4 / Whisper (+ FFmpeg)	Aligned, de-duplicated transcripts
Pre-consultation summarization	Gemini (Flash) + rule-based fallback	Visit update, priority flags, questions
SOAP drafting & symptom extraction	Gemini / Qwen3-8B (optional)	Structured drafts with references
Patient similarity	Weighted Gower + symptom match + cosine + Bayesian shrinkage	Comparable cases with explained matches
Prescription template retrieval	Gower × coverage + template search	Reusable templates with allergy/conflict checks
Workflow automation	Rules-first + Ollama/OpenAI (Gemma 4 E4B)	Validated UI actions (open, fill, save)
Community screening	Local LLM + deterministic rules	Redaction, prioritization and human review

End-to-End Workflow (How It Works)

Patient inputs & clinician actions
Check-ins, uploads, audio, vitals, questions

→

Processing & AI
OCR, transcription, summarization, retrieval

→

Validation & clinician review
Verify, edit and approve notes and prescriptions

→

Patient care plan
Visit summary, prescriptions and follow-up tasks

Integrations & Key Components

Android Health Connect
Steps, sleep, HR, BP, weight

HTTPS Audio Uploads
Retryable chunks (session + device + seq)

Socket.IO
Real-time consultation status

Firebase Cloud Messaging
Community notifications

Email Invitations
Scheduled clinic visits

In-app Follow-up
Check-in prompts & task checklists

Redis (Optional)
Temporary audio caching

Validation & Clinician Control

- **Access** (Role, ownership, RLS)
- **Input** (Schema, file & size)
- **OCR** (Confidence, alias, units)
- **Audio** (Consent, sequence, completeness)
- **AI Outputs** (Structured, cited, traceable)
- **Similarity** (Coverage, shrinkage)
- **Prescription** (Allergy & applicability)
- **Automation** (Allowed actions, confirmation)
- **Community** (Redaction, screening, human review)

Data Sources, Storage & Evaluation

Care Data
Check-ins, vitals, lab reports, maternity records, consultation audio

Storage
Private Supabase buckets (access controlled)

Dev & Evaluation
Synthetic fixtures, curated maternal cases, seeded cohorts

Dataset Support
UCI Maternal Health Risk, MTS-Dialog (compatibility only)

Evaluation: API tests, unit tests and failure drills (uploads, transcripts, retrieval and draft verification).

The platform flags and prioritises; it never diagnoses. Automation requires its feature branch and configuration. Daily scheduled medication push reminders remain planned. Clinical outcome validation is pending.

Build Sprint Journey and Progress

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Health-On progressed from an appointment-preparation and report-digitization concept into an integrated maternal-care prototype. Four development milestones delivered connected records, patient access, consultation documentation, and AI assistance, followed by stronger execution controls and file protection.

Phase 1: Connected Clinic Foundation

Appointments, access, OCR and shared records.

- Auth & role access, appointments & invites.
- Report upload, OCR, normalization & review.
- Notes, prescriptions, visit history & follow-up.

Core foundation integrated

Phase 2: Maternal Records and Patient App

Maternity card, mobile access and screened community.

- Digital maternity card & longitudinal records.
- Investigations, ultrasound, vaccination, etc.
- Patient app (reports, vitals, summaries, tasks).
- Screened peer community & separate identity.

Core maternal-care features integrated

Phase 3: Consultation and Clinical Support

Audio capture, reviewed SOAP drafts and explainable similarity.

- Dual-device recording, reliable uploads.
- Transcription (Sarvam / Whisper) & de-duplication.
- Reviewable SOAP drafts with citations & checks.
- Prescription templates & applicability flags.
- Patient similarity (Gower + symptoms + narrative).

Core pipeline and decision-support interfaces integrated

Phase 4: Continuity, Automation and Protection

Check-ins, wearables, reminders, merged assistant and encryption.

- Prenatal check-ins & pre-visit briefs.
- Health Connect imports & trend views.
- Automation assistant (merged to master) with verification & execution controls.
- AES-256-GCM encryption with RSA-OAEP key wrapping and time-limited signed URLs.

Integration achieved; live acceptance pending

How the Approach Evolved

Initial concept: Appointment preparation and report digitization.

↓

Expanded: Structured maternity records and mobile care access.

↓

Then: Consultation capture and reviewed AI documentation.

↓

Now: Integrated automation with execution verification and added protection for stored files and audio.

Major Challenges and Changes

Challenge	Change made	Progress achieved
Inconsistent reports and uncertain OCR values	Added normalization, verification and clinician review	Established a review path before reuse
Audio boundaries, overlaps and interrupted uploads	Added recorder fixes, alignment, retry handling and gap visibility	Improved processing reliability and failure visibility
Unsupported AI details	Added schema, citation and numeric checks with clinician review	Produced inspectable drafts
Assistant actions could report completion without evidence	Bound results to tasks/records, checked form state and enforced sequential progress	Strengthened execution verification
Automation integration disrupted dashboard behavior	Corrected visit-opening and notes controls while preserving the workspace	Integrated automation into master
Sensitive files required additional protection	Added envelope encryption and expiring signed retrieval	Implemented protection in updated upload and retrieval paths

Remaining Work

- [ ] **Real workflow acceptance:** Test the merged assistant, confirmations, cancellation and multi-step actions in a live browser.
- [ ] **Physical-device validation:** Verify sustained recording, network recovery, Health Connect imports and notification delivery.
- [ ] **Medication-dose reminders:** Prescription-linked daily scheduling remains incomplete.
- [ ] **Audio extensions:** Complete batch transcription cross-check and speaker-labelling.
- [ ] **Encryption deployment:** Validate persistent keys, signed retrieval, recovery procedures and handling of historical files.
- [ ] **Clinical pilot:** Measure accuracy, usability, documentation effort and practical usefulness with clinicians.

Key Capabilities Now Functional

- Appointments & pre-visit preparation
- Digital maternity records
- Lab OCR & standardization
- Consultation capture & transcription
- AI briefs & SOAP drafts (reviewable)
- Patient similarity (with explanation)
- Prescription templates & safety checks
- Prenatal check-ins & wearable trends
- Automation assistant (merged to master)
- Encrypted file storage (AES-256-GCM)

Milestone Outcomes

- A working end-to-end prototype.
- Key clinical, patient and AI workflows integrated.
- Clinician control and review preserved across AI features.
- Foundation ready for stronger automation and protection.

Current position: Core maternal-care workflows and automation are integrated into master. File encryption is implemented. Real-device acceptance, production configuration and clinician pilot evaluation remain ahead.

Safety & Responsible AI

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Core Principle

Health-On flags, summarizes, and supports review. Clinical decisions remain with the clinician.

01 Privacy and Meaningful Consent



- Record consultation consent with its text version and acceptance time.
- Request microphone and Health Connect permissions explicitly.
- Separate community identity from clinical identity and restrict access to authorized users.
- Use generic notification text to avoid exposing clinical details on lock screens.

Consent is recorded, permission-based and access-controlled.

02 Security and Data Protection



- Apply authentication, role and patient-access checks, and database row-level security.
- Store documents and consultation audio in private storage.
- Encrypt new files using AES-256-GCM, with RSA-OAEP protection for file keys.
- Provide time-limited signed retrieval URLs and metadata-focused audio logging.

Encryption is server-side and requires configured deployment keys. Historical plaintext files need separate migration assessment.

03 AI Errors Remain Visible and Reviewable



- OCR, speech and language models can misread, omit context, or generate unsupported statements.
- Route uncertain OCR results for verification and expose incomplete audio or processing gaps.
- Validate draft structure, transcript references, and numeric claims. Flag unreliable content for review.
- Use a deterministic brief fallback when model generation is unavailable (with backend dependencies).

These checks reduce errors but cannot guarantee factual accuracy.

04 Human Oversight and Bounded Automation



- Save AI consultation outputs as drafts for clinician review (edit, approve or reject).
- Keep prescription suggestions editable, with applicability and allergy flags.
- Require confirmation for supported assistant actions that save or send information.
- Check patient context and observed action results; keep recording, consent, destructive actions and file selection manual.
- Route concerning community content to configured professional review and outreach.

AI and automation assist, but final decisions, edits and critical actions remain with the clinician.

05 Bias and Evidence Limits



- Present similarity as descriptive resemblance, with visible matches, conflicts, missing factors and coverage.
- Distinguish missing information from an explicitly denied symptom.
- Treat clinician template choices as preference examples, not proof of treatment effectiveness.
- Synthetic testing and dataset-format adapters do not establish clinical representativeness.
- Evaluate performance across gestational stages, languages, accents, recording conditions and patient groups.

The similarity approach remains an unvalidated retrieval aid.

06 Accessibility and Inclusion



- Offer guided check-ins, visual comfort ratings and short interaction paths.
- Support English, Hindi and Hinglish requests in the website assistant.
- Retain manual reporting and standard navigation for users without wearables or who prefer it.
- Include labeled assistant controls and reduced-motion animation.
- Validate screen-reader use, text scaling, literacy needs and regional-language comprehension with users.

These are accessibility provisions and design intentions; comprehensive accessibility conformance has not been established.

Implemented safeguards support responsible use. Clinical validation, subgroup evaluation, accessibility testing, and deployment checks remain required before rollout.

Deployment, Cost & Sustainability Plan

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Core Approach

Start with a supervised, low-volume pilot using free-tier services and existing hardware. Keep Gemini and Sarvam as usage-based expenses, and distribute future infrastructure costs across participating doctors.

Pilot:
Use free allowances and existing resources where operationally suitable.

Production baseline:
≈ ₹8,100/month for shared infrastructure, plus usage for Gemini, Sarvam and hosted OCR.

Per consultation cost:
≈ ₹13.50–16
(≈ ₹13.48 estimate)

Five-doctor example:
≈ ₹23,000/month
(≈ ₹4,600 per doctor)

01 | Deployment & Pilot Requirements

- Deploy clinician web portal and a shared backend for multiple doctors.
- Use existing laptops, Android phones, microphones and internet connectivity.
- Use Supabase for authentication, clinical records and private file storage.
- Run PaddleOCR and local-model services on existing hardware.
- Configure consent, access permissions, encryption keys, backups and retention procedures.
- Complete synthetic testing, device checks and staff training.
- Assign a clinical lead (AI-output review) and a technical owner (operational issues).

The pilot depends on institutional approval of the workflow, hosting arrangements and external AI processing.

02 | Integrations

Integration	Purpose	Current position
Supabase	Auth, records, private storage	✓ Implemented
Sarvam	Consultation speech-to-text	✓ Provider integrated
Gemini	Briefs, drafts, extraction and automation	✓ Provider integrated
Android Health Connect	Permission-based import of device readings	✓ Implemented
SMTP / Firebase	Invitations and notifications	Configuration & validation required
Hospital EMR / HMIS	Exchange patient, appointment and lab data	Future connector work
ABDM / ABHA & FHIR	Consent-based interoperability	Future integration

03 | Standard Shared Backend Cost (Paid Production Baseline)

Planning conversion: US$1 = ₹90, excluding taxes.

Component	Paid configuration assumed	Monthly cost
Shared Node backend & embedded processing workers	1 CPU, 2 GB RAM — $25	₹2,250
Hosting workspace	$25	₹2,250
Supabase (database, auth, storage)	$25 starting paid configuration	₹2,250
Operational allowance	Bandwidth, monitoring, email and backup expenses	₹1,350
Shared monthly infrastructure budget		**₹8,100**

Published subscription subtotal is ₹6,750; ₹1,350 is a planning allowance. Instance capacity needs load testing. A separate processing worker would add ≈ ₹2,250/month.

04 | Hosted PaddleOCR Pricing

- Baidu Cloud's PaddleOCR-VL document-parsing API (paid option).
- CNY ¥0.05 per successfully processed page.
- ≈ ₹0.63 per page (CNY ¥1 = ₹12.50).
- ≈ ₹625 per 1,000 pages.
- No dedicated OCR inference server in our deployment budget.

≈ ₹0.63 / page
(≈ ₹625 per 1,000 pages)

Excludes free quotas and introductory offers. Requires an adapter and validation (lab extraction, units, confidence). Repo: server/src/services/ocrService.js

05 | Revised Usage Cost Per Consultation

Service	Estimated cost / consultation
Sarvam transcription (≈ 20 audio min)	₹9.80
Gemini processing (40K input, 6K output tokens)	₹2.43
Hosted PaddleOCR-VL (two pages)	₹1.25
Total variable usage cost	**₹13.48**

≈ ₹13.50–16
per consultation

06 | Production Example: Five Doctors (at 220 consultations/doctor/month)

Expense	Monthly estimate
Shared infrastructure budget	₹8,100
Sarvam (1,100 × ₹9.80)	₹10,780
Gemini (1,100 × ₹2.43)	₹2,673
Hosted PaddleOCR-VL (2,200 pages × ₹0.63)	₹1,375
Total estimated production running cost	**₹22,928**
Allocated cost per doctor	**₹4,586 / month**
Allocated cost per consultation	**₹20.84**

- 1,100 consultations (5 × 220) and 2,200 OCR pages.
- Operating-cost estimates, before taxes and commercial margins.
- Excludes salaried support, clinical moderation, custom integrations and HA infrastructure.

07 | How Shared Costs Reduce Per Doctor

$$\text{Monthly cost per doctor} = \frac{₹8{,}100}{\text{number of doctors}} + (\text{monthly consultations per doctor} \times ₹13.48)$$

Doctors sharing the system	Facility running cost / month	Allocated cost per doctor / month
1	₹11,066	₹11,066
5	₹22,928	₹4,586
10	₹37,756	₹3,776
20	₹67,412	₹3,371

The larger rows illustrate cost sharing within an unchanged infrastructure budget. Actual compute and provider plans must increase when concurrency or capacity requires it.

08 | Sustainability & Long-Term Plan

- Measure audio minutes, tokens, storage and delays.
- Upgrade shared services when quotas require.
- Move selected workloads to local models (e.g., Gemma, Qwen, Whisper) after validation.
- Retain clinician review and community moderation.
- Expand across facilities after validating isolation and peak use.
- Fund via facility subscriptions, with institutional partnerships for early pilots.

Initial pilot: free-tier infrastructure and existing hardware, with ≈ ₹12–15 in AI usage per standard consultation. Future operating costs are shared across doctors, while validated local models provide a pathway to reduce API dependence.

What Next? Plans to Take the Initiative Forward

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Core Objective | Validate the solution in real clinical settings, expand capabilities, enable interoperability, and build a sustainable, responsible deployment.

01 Pilot, Validate and Build Partnerships



- Partner with an antenatal clinic or PHC for a supervised pilot.
- Begin with a small clinician group and expand after reviewing feedback.
- Measure documentation time, clinician corrections, OCR/transcription quality, patient participation, and cost per consultation.
- Use findings to prioritize improvements before wider deployment.
- Seek clinical collaborators, hospital-software partners, and incubation or funding support.

02 ABDM Integration and Verified Access



- Integrate ABHA creation/verification and patient-record linking.
- Enable consent-based retrieval of ABHA-linked records.
- Explore Healthcare Professionals Registry (HPR) and Health Facility Registry (HFR) integration for verification.
- Complete the applicable sandbox, functional-testing, and security-review process.

An ABHA number alone does not authorize access to a patient's records; access requires patient consent (ABDM).

03 Expanded Roles and Team Workflows



- Add receptionist workflows for registration, appointments, and queues.
- Introduce nurse/antenatal-care staff workflows for intake, vitals, and preparation.
- Add facility-administrator controls for staff onboarding and access management.
- Define permissions for each role, preserving clinician authority over clinical documentation.
- Verify non-clinical staff through facility-managed onboarding.

04 Patient Similarity and Limited Predictions



- Expand the similarity engine for more verified longitudinal records and different data formats.
- Improve missing-data handling, confidence reporting, and explanations of matches/conflicts.
- Define specific prediction targets (e.g., follow-up needs or care-task completion) before development.
- Evaluate predictions alongside the existing explainable similarity baseline.
- Require clinician review, calibration, subgroup testing, and independent validation before clinical use.

05 Interoperability Adapter for Other Software



- Create a documented API and adapter toolkit for hospital, laboratory, and healthcare software.
- Accept structured patient records, reports, and visit information through authenticated interfaces.
- Map patient identifiers, units, timestamps, and source information into a common format.
- Validate imports, detect duplicates, and preserve provenance.
- Support standards-based exchange, including relevant FHIR profiles, as integrations mature.

06 Patient Follow-Up and Inclusion



- Add medication reminders based on clinician-approved prescriptions.
- Improve task completion tracking and consented follow-up prompts.
- Extend regional-language support and test mixed-language consultation speech.
- Validate screen-reader access, text scaling, and usability with patients of different literacy levels.
- Improve recovery from interrupted uploads and weak connectivity.

Make the solution publicly available? | **Yes**

We plan to share the solution, documentation, and key findings (including known limitations) on the Health-a-thon platform and an open project site, following the pilot.